

Assets Monitoring System

Project Description

Project Owner	Mahmoud Morshedy
Role	BI Engineer
Project Type	Self-initiated internal BI solution
Tools Used	Power BI · DAX · Power Query · SharePoint Online
Status	Live / In active use

Project Overview

The Assets Monitoring Dashboard is a self-initiated Power BI solution built to give IT operations team a single, reliable source of truth for hardware asset tracking. Prior to this project, hardware data — including device assignments, lifecycle stage, location, and maintenance history — was scattered across multiple SharePoint lists and manually maintained spreadsheets, making it difficult to answer simple questions like “how many laptops are available in our stock room right now?” without a manual audit.

By connecting directly to SharePoint Online and modeling the data in Power BI, the dashboard consolidates asset records into a live, interactive view that updates automatically as records change — removing the need for manual reconciliation and giving stakeholders real-time visibility into the hardware fleet.

Objectives

- Centralize hardware asset data into a single, governed source of truth.
- Provide real-time visibility into asset allocation and location.
- Reduce time spent on manual audits and spreadsheet reconciliation.
- Give IT leadership a clear, at-a-glance view of fleet health and utilization.

Data Source & Architecture

The dashboard pulls data directly from SharePoint Online lists using Power Query, which handles extraction, cleaning, and transformation before the data is loaded into the Power BI data model. Key architectural decisions include:

- **Live connection:** Direct SharePoint Online connector ensures the dashboard reflects the current state of records without manual refresh of source files.
- **Data modeling:** A snowflake-schema-style model separates asset facts from dimension tables enabling efficient filtering and aggregation.
- **DAX measures:** Custom DAX calculations power KPIs such as asset count by status.
- **Data quality rules:** Power Query transformation steps standardize inconsistent entries (e.g. naming conventions for device types) before they reach the report layer.

Key Features

- **Asset lifecycle tracking:** Visual breakdown of hardware by lifecycle stage (new, in-use, used).
- **Allocation overview:** Distribution of assets across departments and locations, filterable by device type.
- **Interactive filtering:** Slicers for manufacturer, device type, and status allow stakeholders to drill into the specific slice of the fleet they manage.
- **Summary KPI cards:** At-a-glance metrics for total assets and total in stock assets.

Impact

Since deployment, the dashboard has replaced what was previously a manual, spreadsheet-driven audit process with a live reporting layer that any stakeholder can access on demand. This has reduced the time spent compiling hardware status reports and has improved the team's ability to plan order budgets ahead rather than reacting after failures occur.

Skills Demonstrated

- Power BI report design and dashboard layout
- DAX measure development for dynamic KPIs
- Power Query (M language) for data extraction and transformation
- SharePoint Online as a live data source
- Data modeling best practices (snowflake schema, dimension/fact separation)
- Stakeholder-driven requirements gathering and iterative design